



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES -(24SDCS03)

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TITLE OF THE PROJECT	SymptoSense AI: A Multi-Agent Health Symptom Analytics, Risk Prediction, and Care Routing Ecosystem	
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Abstract:

Emergency rooms around the world are overcrowded, and many of the patients who show up don't actually need emergency-level care — they could have been checked earlier, given advice, or sent somewhere more appropriate. Right now, this doesn't happen well because current symptom-checker apps are basic: they follow fixed yes/no question trees, can't handle people describing multiple confusing symptoms at once, and don't explain why they gave a certain suggestion. This makes both patients and doctors unable to trust them.

Because of this, there's a real need for a smarter system — one that can understand what a patient is saying in plain language, figure out how serious their condition might be, and then guide them to the right kind of care.

SymptoSense AI is built to do exactly that. Instead of one single program trying to do everything, it uses three smaller AI "agents" that each handle one job and work together.

The first agent- Symptom Analytics performs symptom analysis on text input from the patient, using GPT-4o/Gemini to extract key medical details and convert them into structured data for further processing..

The second agent -Risk Prediction takes that information and predicts how risky or serious the situation is — and importantly, it explains its reasoning instead of just giving a mysterious score.

The third agent-Care Routing then decides what should happen next: send the patient to the ER if it's truly urgent, book a video doctor visit for moderate cases, or just give safe self-care advice for minor issues. This keeps emergency rooms free for people who actually need them.

Collectively, these three agents form an autonomous, explainable, and safety-conscious triage ecosystem that shifts symptom checking from static lookup to genuine multi-agent reasoning. By combining analytics, predictive risk scoring, and intelligent care routing within a service-oriented architecture, SymptoSense AI aims to reduce preventable delays in care-seeking behavior, ease unnecessary burden on emergency systems, and improve early access to appropriate care.

Ultimately, SymptoSense AI is not intended to replace doctors or provide a definitive diagnosis. Its purpose is to act as an intelligent first-level triage assistant that helps users understand the urgency of their symptoms and identify the most appropriate next step in care. This directly supports SDG 3: Good Health and Well-Being.

Project Architecture/Project Overview:

